

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Carbon and its Compounds

CBSE Science 2026-27 | Board-examinable

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

Why does carbon form a very large number of compounds?

Answer:

Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Its ability to form chains, branches, and rings is called catenation.

Question 2

Write the reaction for complete combustion of methane and name the products.

Answer:

$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Complete combustion in sufficient oxygen produces carbon dioxide and water.

Question 3

Why do soaps form scum in hard water, while detergents work better?

Answer:

Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. Detergents do not form these insoluble salts as readily, so they clean effectively in hard water.

Question 4

What is the central idea or skill tested in Carbon and its Compounds?

Answer:

The central focus is: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Carbon and its Compounds.

Answer:

First point: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements.

Second point: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Carbon and its Compounds lead to an important result or conclusion?

Answer:

Begin with the key idea: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Then connect it to the result: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$.

Carbon and its Compounds - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Carbon and its Compounds?

Answer:

Use this supporting detail: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Its ability to form chains, branches, and rings is called catenation. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Carbon and its Compounds.

Answer:

$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Complete combustion in sufficient oxygen produces carbon dioxide and water. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Carbon and its Compounds?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather.

Question 10

Compare the first and second important ideas in Carbon and its Compounds.

Answer:

The first idea is: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. The second idea is: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Carbon and its Compounds.

Answer:

Cause or starting point: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Effect or outcome: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Carbon and its Compounds?

Answer:

A clear sequence is: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Next, $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Finally, Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather.

Carbon and its Compounds - Continued

Question 13

Give a 40-50 word answer outline for an important question from Carbon and its Compounds.

Answer:

Start with the direct answer: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Add one supporting point: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. End with the outcome or message: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Carbon and its Compounds?

Answer:

Use an accurate scientific statement: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Carbon and its Compounds?

Answer:

Connect the idea to its reason or result. For instance: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Carbon and its Compounds.

Answer:

A sound inference is that soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Carbon and its Compounds.

Answer:

Apply the concept by first stating: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Then justify the response with: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$.

Question 18

What would make an answer from Carbon and its Compounds incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Also add: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather.

Carbon and its Compounds - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Carbon and its Compounds.

Answer:

Assertion: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements.

Reason: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Carbon and its Compounds in two clear steps?

Answer:

Step 1: state the concept or event - Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Step 2: support it with the explanation or result - $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$.

Question 21

What answer structure is suitable for a 3-mark question from Carbon and its Compounds?

Answer:

Use three connected points: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Carbon and its Compounds?

Answer:

Give an introduction, develop these key points - Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. - and close with: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Carbon and its Compounds be applied to a new situation or case study?

Answer:

Identify the relevant principle first: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Then use the chapter's reasoning to explain the case: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$.

Question 24

Which detail from Carbon and its Compounds is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. Explain it in context instead of using it as an isolated statement.

Carbon and its Compounds - Continued

Question 25

Write a brief concluding statement for an answer on Carbon and its Compounds.

Answer:

A suitable conclusion is: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for Carbon and its Compounds.

Answer:

1. Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. 2. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. 3. Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Carbon and its Compounds before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. - and supported it with a reason, example, calculation, or outcome such as: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$.

Question 28

Why does Carbon and its Compounds remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather.

Question 29

Create a final 60-80 word answer plan for a long question from Carbon and its Compounds.

Answer:

Opening: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. Development: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$. Conclusion: Soap reacts with calcium and magnesium ions to form insoluble scum, reducing lather. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Carbon and its Compounds?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: Carbon is tetravalent and forms strong covalent bonds with itself and many other elements. and $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.